

population parameters

The diagram illustrates the relationship between population parameters and the components of a mixed-effects model equation. At the top, the text "population parameters" has two blue arrows pointing downwards to the intercept and slope terms of the equation below. The equation is $y_{ij} = \alpha + a_j + X(\beta + b_j) + \epsilon_{ij}$. A blue bracket under $\alpha + a_j$ is labeled "varying intercept", and a blue bracket under $\beta + b_j$ is labeled "varying slope".

$$y_{ij} = \underbrace{\alpha + a_j}_{\text{varying intercept}} + X(\underbrace{\beta + b_j}_{\text{varying slope}}) + \epsilon_{ij}$$

varying intercept

varying slope